by an approximate series expansion. In modern computer architectures, the number of force evaluations is responsible for the overwhelming majority of the time spent per iteration.

We can apply the splitting approach considered in Chap. 7 to integrate equations (8.27)–(8.29). One suitable choice of algorithm that we shall use in the following subsection is

$$
\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\xi_{n+1/2} &= \xi_n + (h/2)\mu^{-1}\left[\boldsymbol{p}_{n+1/2}^T\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2} - \mathrm{N_d k_B T}\right],\\
\tilde{\boldsymbol{p}}_{n+1/2} &= e^{-(h/2)\xi_{n+1/2}}\boldsymbol{p}_{n+1/2},\\
\hat{\xi}_{n+1/2} &= e^{-\gamma h}\xi_{n+1/2} + \zeta R_n,\\
\hat{\boldsymbol{p}}_{n+1/2} &= e^{-(h/2)\hat{\xi}_{n+1/2}}\tilde{\boldsymbol{p}}_{n+1/2},\\
\xi_{n+1} &= \hat{\xi}_{n+1/2} + (h/2)\mu^{-1}\left[\hat{\boldsymbol{p}}_{n+1/2}^T\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2} - \mathrm{N_d k_B T}\right],\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\boldsymbol{p}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),
\end{aligned}
$$

where $\zeta = \sqrt{k_\mathrm{B}\mathrm{T}\mu^{-1}(1-\exp(-2\gamma h))}$ and R_n is a normal random number with unit variance and zero mean.

This numerical method requires only one evaluation of the force per timestep. It also reduces to a Nosé-Hoover integrator when $\gamma = 0$, and as it solves the Ornstein-Uhlenbeck process exactly, it is stable under the large (or infinite) limit of friction.

8.4.3 Numerical Comparison of Dynamics

We shall compare the long-time averages evaluated by computing solution trajectories for Nosé-Hoover, Langevin and Nosé-Hoover-Langevin dynamics. We will look at the impact that the choice of dynamics (and associated parameters) has on the rate of convergence of averages in time, as well as the perturbation introduced into dynamical quantities (such as autocorrelation functions).